

Shreyas Wankhede

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Experience

University of Toronto - Mechanical Research Assistant

Nov 2024 – Aug 2025

- Developed fiber-reinforced tie-layers to improve PPS adhesion, achieving 86% peel strength and a 2× mechanical gain by implementing advanced composite techniques.
- Characterized 15+ materials using DMA, TMA, and Rheometry, correlating processing parameters with composite performance to enhance material durability.
- Molded and prepared 30+ composites using three advanced processing methods, resulting in a ~90% increase in durability and performance.
- Collaborated with a team to optimize composite processing, reducing material waste by 22% & improved efficiency.

Bindwel & Stelda (Impel Services Pvt. Ltd.) - Technical Process Intern

May 2023 – April 2024

- Designed a custom packaging system in SolidWorks for 14 SKUs, improving adaptability, optimizing workflow, and reducing setup time by 25%
- Created AutoCAD process sheets that reduced assembly errors by 35% and enhanced manufacturing consistency.
- Awarded a Pre-Placement Offer for exceeding design expectations and accelerating prototype readiness by 15%.
- Implemented process improvements that led to a ~20% increase in production efficiency and a noticeable reduction in overall operational costs

Certifications

Certified SOLIDWORKS Associate (CSWA) - [Certificate](#)

- Gained proficiency in CAD, mastering the fundamentals to create complex assemblies and detailed drawings.

GD&T, Stack-up, ASME Y14.5-2009, Tolerance, GDTP - [Certificate](#)

- Acquired skills of geometric dimensioning & tolerancing principles, precision engineering & manufacturing processes.

Lean Six Sigma Foundations - [Certificate](#)

- Mastered the six principles, process improvement efficiency to implement quality management strategies.

Leadership & Projects

Range-Controlled Ball Launcher Design - [Project](#)

- Led the design and development of a spring-actuated launcher, enhancing precision and reliability through iterative design and testing, and learned to optimize engineering solutions using SolidWorks and Arduino.

Energy-Harvesting System for Gym Equipment - [Project](#)

- Designed a motion-to-electricity conversion system, improving energy efficiency and sustainability, and gained expertise in mechanical design and simulation using SolidWorks and ANSYS.

Education

University of Toronto - Master of Engineering, Mechanical and Industrial Engineering

Indian Institute of Technology Bombay - Bachelor of Technology, Mechanical Engineering

Courses: Product Design, Advanced Manufacturing Technologies, MEMS Design & Microfabrication, Applied CFD

Concentration: Advanced Manufacturing & Product Design, Machine Design, Design for Manufacturing

Skills

Technical Skills: Advanced composites and fiber-reinforced polymers, Product and mechanical design, Manufacturing and fabrication processes, Finite element analysis (FEA), Computational fluid dynamics (CFD), Tolerance and stack-up analysis, Geometric dimensioning and tolerancing (GD&T), Engineering project management.

Software Knowledge: SolidWorks, AutoCAD, ANSYS, ABAQUS, COMSOL, UG NX, Minitab, Arduino programming, Microsoft Office Suite, Notion.

Soft Skills: Problem-solving, Team collaboration, Analytical thinking, Communication, Attention to detail, Innovation.

Volunteer Experience

Crescent 3D Printing - 3D Printing/Modelling Workshop Instructor

Nov 2025 – Present

- Conduct virtual 3D modelling and printing workshops for students (Grades 2 to 12) from Canada, the US, and China, fostering creativity and STEM learning through beginner-friendly design tools.